INTERNATIONAL UNIVERSITY OF APPLIED SCIENCES

ORAL PROJECT REPORT

Task for Course: DLBCSEESE02_E – Project: Social Engineering

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1. TASK

For the oral project report, you can choose from the following tasks. Please choose one of them that you would like to work on in your oral project report.

Note on copyright:

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1.1 Task 1: How Employees Can Protect Themselves From Social Engineering Attacks

Background: You have been tasked with training the staff of a company in the best ways to protect themselves from social engineering attacks. The company that hired you recently fell victim to a social engineering attack. You will analyze this attack and incorporate it into your training.

Here is what happened in the company: An attacker contacted an employee named Peter via e-mail. In the e-mail, the attacker posed as a member of the IT department and claimed there was a problem with Peter's password. He then asked Peter to reset his password. Concerned, Peter followed the instruction and provided his password to the attacker. The attacker used the password to gain access to the company's network. The attacker used the network access to steal sensitive information, including customer data, financial data, and trade secrets. He then sold the information to a competitor of the company. The competitor used the information to deceive the company's customers, offering them products or services at a lower price than the company. The company suffered a financial loss of one million euros and had to pay customer compensation and improve security measures.

Task:

1. Analyze the scenario and identify the various social engineering techniques used by the attacker.
2. Create a presentation explaining these techniques and how employees can protect themselves from such attacks.

1.2 Task 2: Compliance and Code of Conduct: An Effective Protection Against Social Engineering Attacks

Background: Anna is a young employee in the marketing department of a large company. She is ambitious and wants to advance in her career. One day, she receives an email from an unknown sender. The email asks Anna to share confidential information about new products of the company with the sender. Anna is alarmed. She knows she must not disclose confidential information. She remembers the company's compliance and code of conduct. Anna refuses to share the information and reports the incident to her supervisor.

Her supervisor is alarmed and initiates an investigation. He finds that the attacker is an insider within the company. He works in the marketing department and has access to confidential information. The attacker is dismissed, and the incident is reported to the police. Thus, the compliance and code of conduct have proven to be a successful means of protecting the company.

You were responsible for creating the code at that time. In a training session for new employees, you introduce the guidelines and raise awareness about this issue.

Task:

1. Many cyber threats faced by enterprises stem from employees' lack of awareness or insufficient training on cybersecurity. In your presentation, emphasize the importance of recognizing social engineering tactics and provide guidance on how to respond effectively in challenging situations.

2. Develop a comprehensive compliance and code of conduct framework designed to safeguard employees from social engineering attacks. Present this framework in your presentation.

Introductory Literature References:

Kotkova, B., & Hromada, M. (2021). The threat of social engineering and the safety of companies. In *2021 25th International Conference on Circuits, Systems, Communications and Computers* (pp. 126–133). IEEE.

Musil, J. P. (2022). Conducting a cybersecurity compliance review: A how-to guide. *Benefits Magazine, 59*(4), 20–26.

Steinmetz, K. F., Holt, T. J., & Brewer, C. G. (2023). Developing and implementing social engineering-prevention policies: A qualitative study. *Security Journal, 2023,* 1–19.

1.3 Task 3: Social Engineering: International Collaboration Prevents Data Breach

Background: You have been tasked with training company executives on how best to protect a company from social engineering attacks. The company that hired you recently fell victim to a social engineering attack, where international collaboration was able to prevent a data breach. You will analyze this attack and incorporate it into your training.

An employee of a German company receives an email from a supposed customer in the USA. The email contains a link to a document that urgently needs to be read. The employee clicks on the link and is redirected to a fake website of the company. On the website, he is asked to enter his login credentials. The employee enters his credentials and is then redirected to another fake website. On this website, he is asked to enter sensitive data. The employee becomes suspicious and reports the incident to his IT department. The IT department informs the national cybersecurity authority. The cybersecurity authority collaborates with the FBI. The authorities exchange information and work together on an investigation. The collaboration between the authorities leads to the identification and arrest of the attacker. The employee can reset his credentials, and the stolen data is deleted.

Task:

Please prepare a presentation that addresses the following questions:

1. How can national and international collaboration help in combating social engineering?
2. What potential impacts can social engineering attacks have on data breaches?
3. What comprehensive measures can companies take to protect themselves from social engineering attacks that can lead to data breaches?

Introductory Literature References:

Hadnagy, C. (2010). *Social engineering: The science of human hacking.* Syngress.

Mishra, A. Alzoubi, Y. I., Gill, A. Q., & Anwwar, M. J. (2022). Cybersecurity enterprises policies: A comparative study. *Sensors, 22*(2), 538. 2022.

Takahashi, T., Kadobayashi, Y., & Nakao, K. (2011). Toward global cybersecurity collaboration: Cybersecurity operation activity model. In *Proceedings of ITU Kaleidoscope 2011: The Fully Networked Human? Innovations for Future Networks and Services* (pp. 1–8). IEEE.

2. ADDITIONAL INFORMATION FOR THE EVALUATION OF THE ORAL PROJECT REPORT

When conceptualizing the oral project report, the evaluation criteria and explanations given in the writing guidelines should be considered.

3. TUTORIAL SUPPORT

Several options are available for support with presentations. The student is responsible for making use of these resources. Tutors are available for subject consultation on the choice of topic as well as for specific and general questions on academic work. There is no provision for the tutor to confirm acceptable outlines, parts of the content, or presentation drafts, since independent preparation is part of the examination. However, hints may be given on rough drafts to facilitate the creation of academic work.